NETWORK & SECURITY ENGINEER

Hassan Ismail.

CS & AI Student, South Valley National University

Working across two tracks: competitive programming in C++ (ECPC/ICPC) and network security.

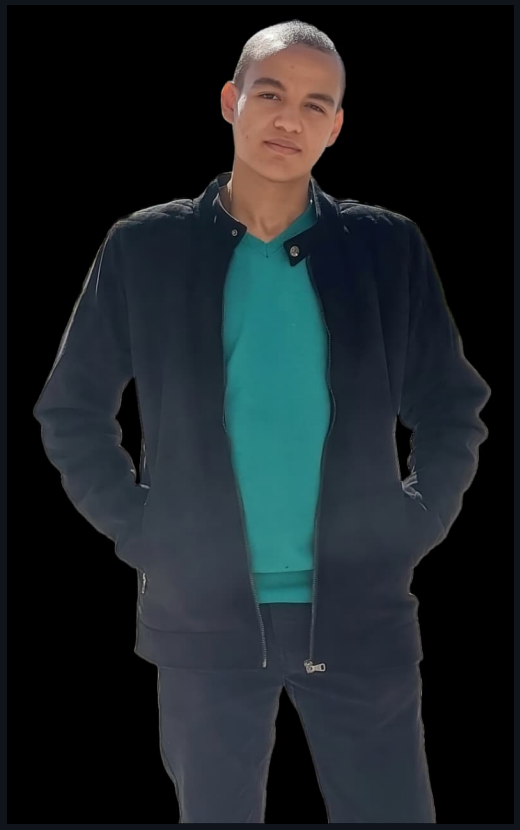
> “Failure is a milestone for success.”

Hassanismailtheemperor@gmail.com

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LinkedIn Profile

Qena, Egypt



ID: H.ISMAIL

AVAILABLE

SELECTED WORK

01 **HA Cluster <-> SD-WAN Site-to-Site VPN**
Two FortiGate sites joined over an encrypted tunnel

02 **DMZ Segmentation**
Three-legged firewall isolating public-facing services

03 **Firewall Policy Engine**
Firewall rule-matching logic, rebuilt from scratch in C++

SKILLS & TOOLS

What's actually behind the work

Trained hands-on through NTI's FortiOS Administrator program and independent practice.

CORE SKILLS

- Firewall Policy Design
- High Availability (HA) Clustering
- Site-to-Site IPsec VPN
- SD-WAN (Multi-WAN)
- Network Segmentation (DMZ)
- Static & Failsafe Routing
- CLI-Based Verification & Diagnostics

TOOLS & PLATFORMS

- FortiOS / FortiGate VM64
- FortiGate CLI Console
- Python (Intro + Intermediate)
- Network Fundamentals & Protocols

EDUCATION & TRAINING

Currently building the fundamentals

Second-year Computer Science & AI student, with applied network-security practice and two completed Python courses alongside coursework.



ENTERING YEAR 3

South Valley National University

Faculty of Computer Science & Artificial Intelligence - CS & IT

Completed the second year, now entering the third. Coursework in programming, algorithms, and computer networks runs alongside the independent FortiGate work in this book.



NTI - 2026

National Telecommunication Institute

FortiOS Administrator -- Network Security, Aug 4-31, 2026

Official Fortinet Academy curriculum (60 technical hours): firewall policies, the Security Fabric, authentication, and security profiles (IPS, antivirus, web filtering, app control), plus IPsec VPN, SD-WAN, and HA -- the practical basis for the FortiGate projects in this book, built toward the NSE4 FortiOS Administrator path (exam NSE4_FGT_AD-7.6). The program also included a 12-hour freelance module on packaging those skills into client-ready services. Also completed two Microsoft-backed Python courses under an MCIT digital-skills program.



Network Security
Aug 4-31, 2026



Intro to Python
Sep 14-23, 2025



Python - Intermediate
Sep 25-Oct 2, 2025

COMPETITIVE PROGRAMMING

C++ contest record

Alongside network security, I compete in C++ with South Valley National University's ECPC/ICPC team.

APR 2025 - INDIVIDUAL

ICPC ACPC Kickoff
Online Individual Contest

Honorable Mention


Certificate of Achievement

awarded to

Hassan Ismail Hassan Mohamed

South Valley National University

The 2025 ICPC ACPC Kickoff Online Individual Contest
25 - 26 April 2025

Honorable Mention

William B. Poucher, Ph. D.
ICPC Executive Director

AUG 2025 - PERSONAL CERTIFICATE

ECPC Qualifications
Collegiate Prog. Contest, Day 8

94th Place


Certificate of Achievement

awarded to

Hassan Ismail Hassan Mohamed

South Valley National University

The 2025 ICPC ECPC Qualifications Collegiate Programming Contest Day 8
07 - 07 August 2025

Ninety-fourth Place

William B. Poucher, Ph. D.
ICPC Executive Director

AUG 2025 - TEAM CERTIFICATE

ECPC Qualifications
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94th Place


Certificate of Achievement

The 2025 ICPC ECPC Qualifications Collegiate Programming Contest Day 8
07 - 07 August 2025

Ninety-fourth Place

South Valley National University

Hassan Ismail Hassan Mohamed
Khaled Magdy Wahib
Mostafa mohammed abo el hassan
Mohamed Rady, Coach
Maram Mahmoud Mostafa, Co-coach

William B. Poucher, Ph. D.
ICPC Executive Director

COMPETITIVE PROGRAMMING -- CONTINUED

AUG 2026 - PERSONAL CERTIFICATE

ECPC Qualifications
Collegiate Prog. Contest, Day 8

93rd Place



Certificate of Achievement

icpc.foundation
icpc International Collegiate Programming Contest

awarded to

Hassan Ismail Hassan Mohamed

South Valley National University

The 2026 ICPC ECPC Qualifications Collegiate Programming Contest Day 8
20 August 2026

Ninety-third Place

William B. Poucher, Ph. D.
ICPC Executive Director

AUG 2026 - TEAM CERTIFICATE

ECPC Qualifications
Collegiate Prog. Contest, Day 8

93rd Place



Certificate of Achievement

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The 2026 ICPC ECPC Qualifications Collegiate Programming Contest Day 8
20 August 2026

Ninety-third Place

South Valley National University

Hisham Ali Mohamed Elsadek
Hassan Ismail Hassan Mohamed
Mostafa mohammed abo el hassan
Mohamed Hassan Esmael, Co-coach

William B. Poucher, Ph. D.
ICPC Executive Director

WHAT I DO

Services

Scope drawn directly from the work in this book.

01

Firewall & Policy Design

Rule sets built around least privilege - explicit deny where it matters, scoped ACCEPT rules everywhere else, all under UTM inspection.

Project 01 - 02

02

High Availability

Active-passive FortiGate clustering with unit priority and synchronization, so a single hardware failure doesn't take the network down.

Project 01

03

Site-to-Site VPN

IPsec tunnels between independent sites, with routing that prefers the tunnel and falls back to a blackhole route if it drops.

Project 01

04

SD-WAN

Multi-WAN edges grouped into a single zone, so outbound traffic keeps moving even if one link degrades.

Project 01

05

Network Segmentation

Zone-based design - LAN, DMZ, WAN - with firewall policy controlling what can reach what.

Project 02

06

Verification & Testing

Every path is confirmed from the CLI - sourced pings, tunnel status, session counters - before it's called done.

Project 01 - 02

PROJECT 03

Firewall Policy Engine

Real network security work above; this is the logic underneath it, rebuilt from scratch in C++ -- not configured through a GUI, but implemented.

Loads ACL-style policy from a file and evaluates packets against source/destination subnet -- real CIDR bitmask matching, not string comparison -- plus protocol and port range. Rules are checked top to bottom, first match wins, with a default implicit deny if nothing matches. A trace mode walks every rule against every packet and reports exactly which condition caused each one to match or fail.

`Language: C++` `Matching: CIDR / Protocol / Port` `Verification: Per-rule trace mode`



github.com/themodsstealer/Firewall-policy-engine

PROJECT 01

HA Cluster <-> SD-WAN Site-to-Site VPN

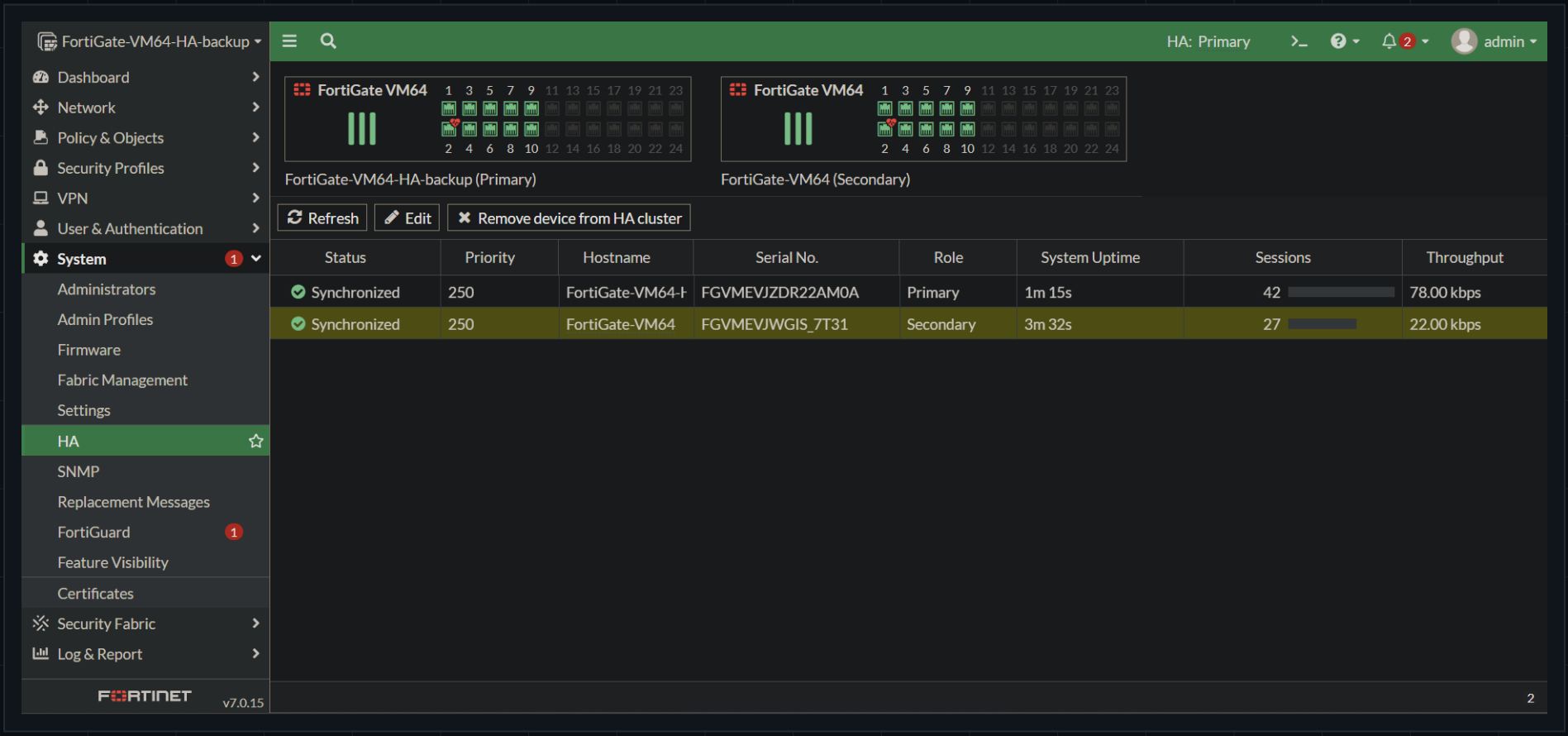
P1 runs two FortiGate VM64 units in an HA cluster (priority 250, both synchronized) fronting the 192.168.10.0/24 LAN. P2 runs a single FortiGate with two WAN links bonded into an SD-WAN virtual-wan-link, fronting the 192.168.20.0/24 LAN. A site-to-site IPsec tunnel connects the two, with static routes on both sides preferring the tunnel and falling back to a blackhole route so traffic never leaks out to the internet if the VPN drops.

Role: Primary / Secondary	Priority: 250	Tunnel: Up - both directions	Verification: CLI ping, 0% loss

SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up



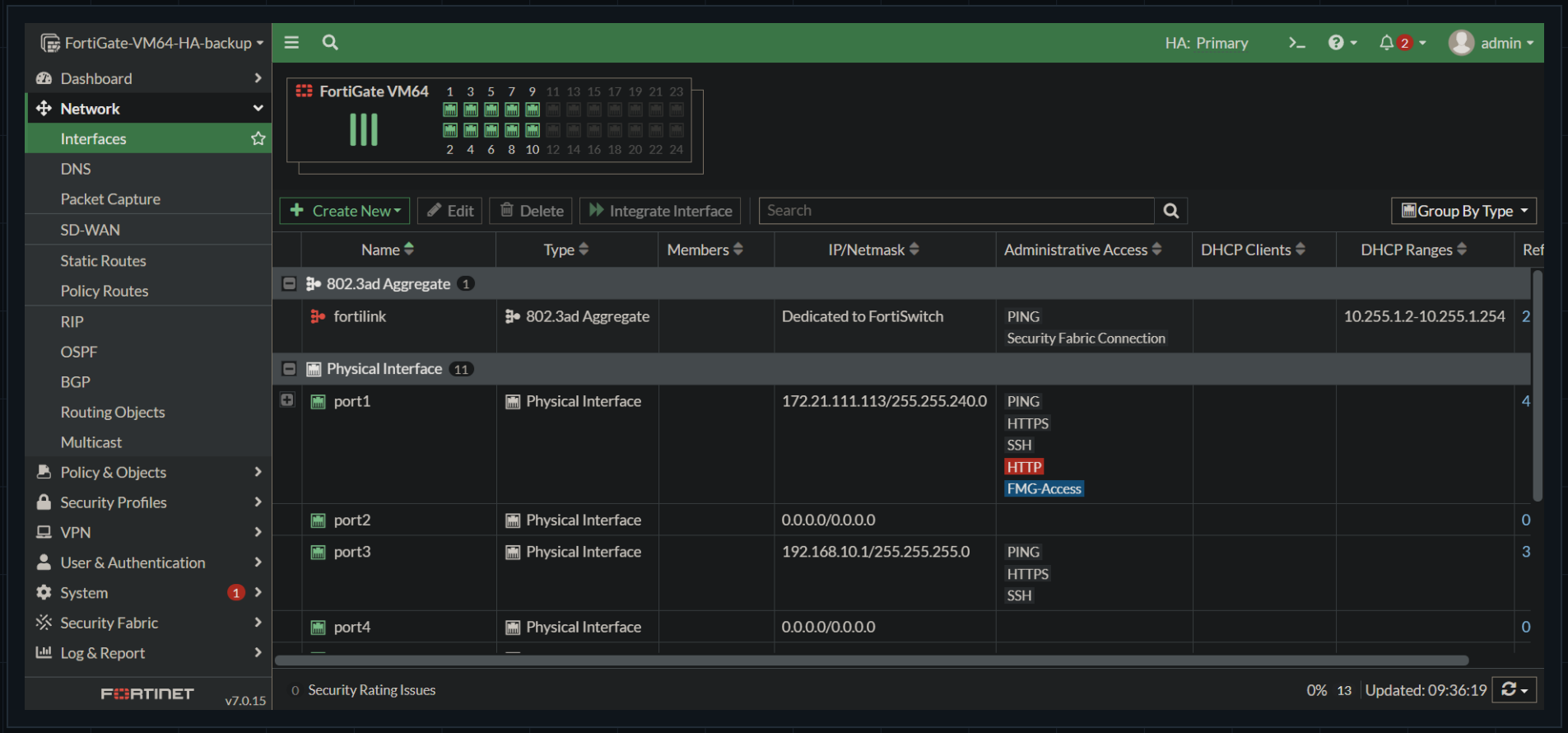
01 HA Status

Active-passive HA pair, both synchronized at priority 250 - live session count and throughput per unit.

SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up



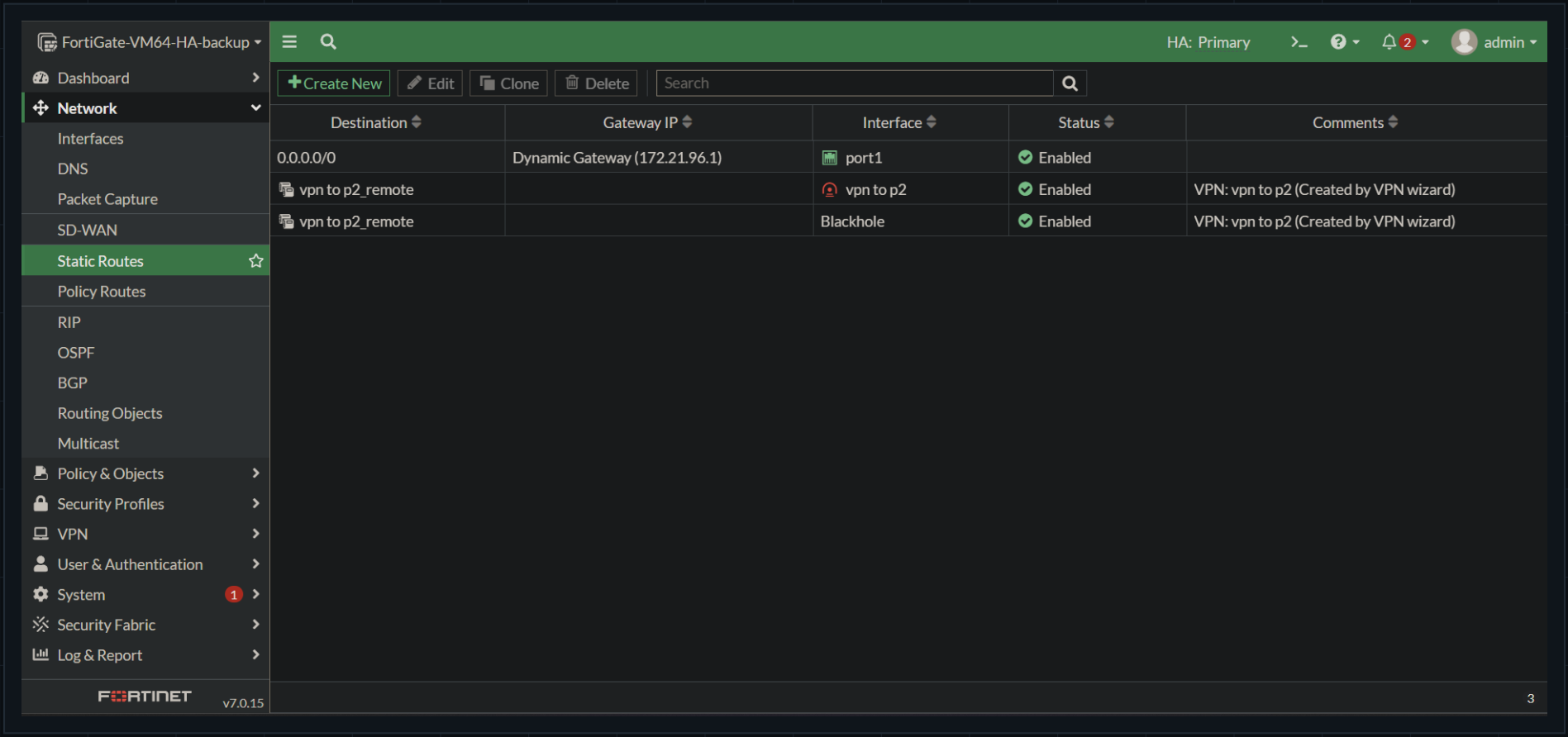
02 Interfaces

port1 carries WAN/management (SSH, HTTPS, FMG-Access); port3 terminates the internal 192.168.10.0/24 LAN.

SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up



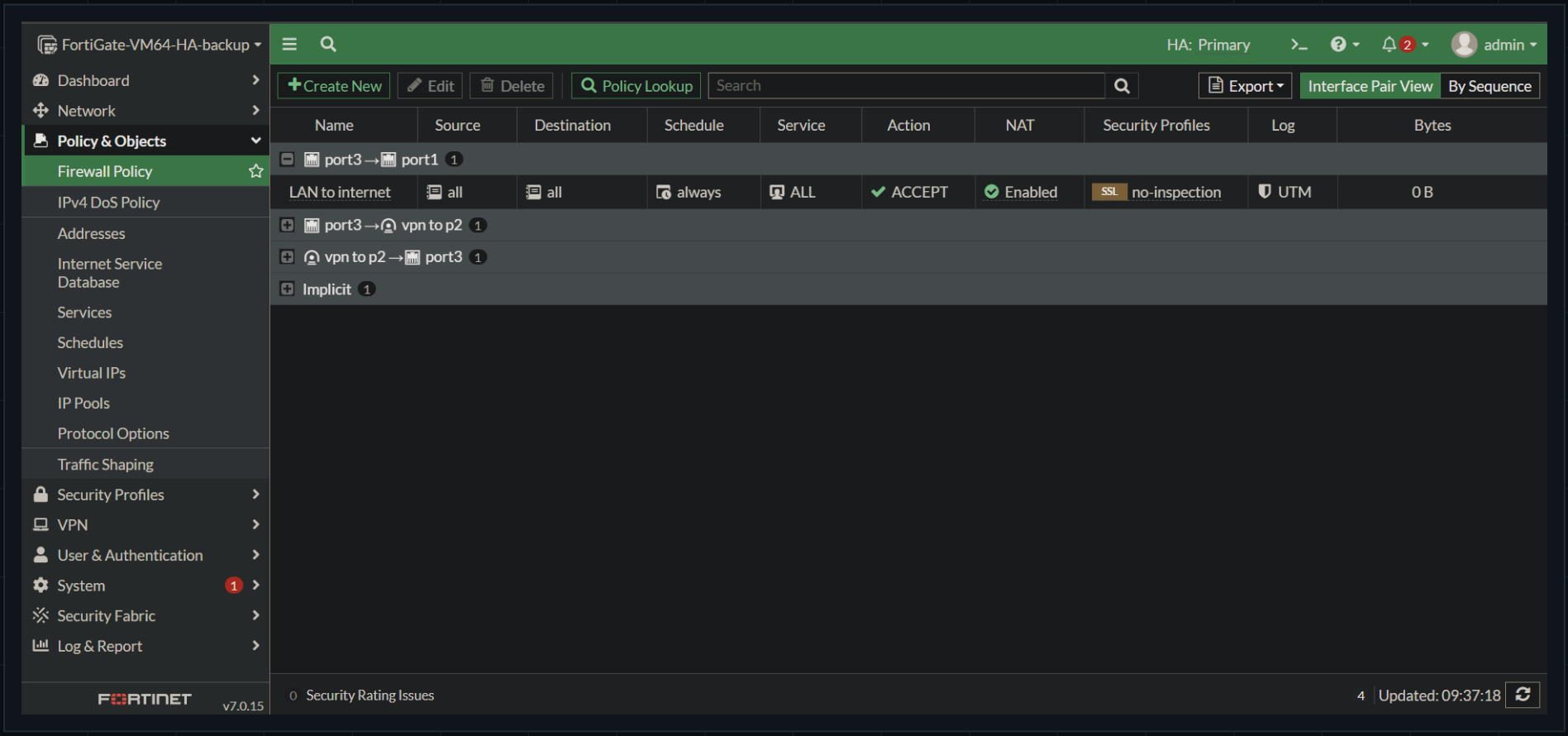
03 Static Routing

Default route out port1, a primary route through the tunnel, and a blackhole route so nothing leaks if the VPN drops.

SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up

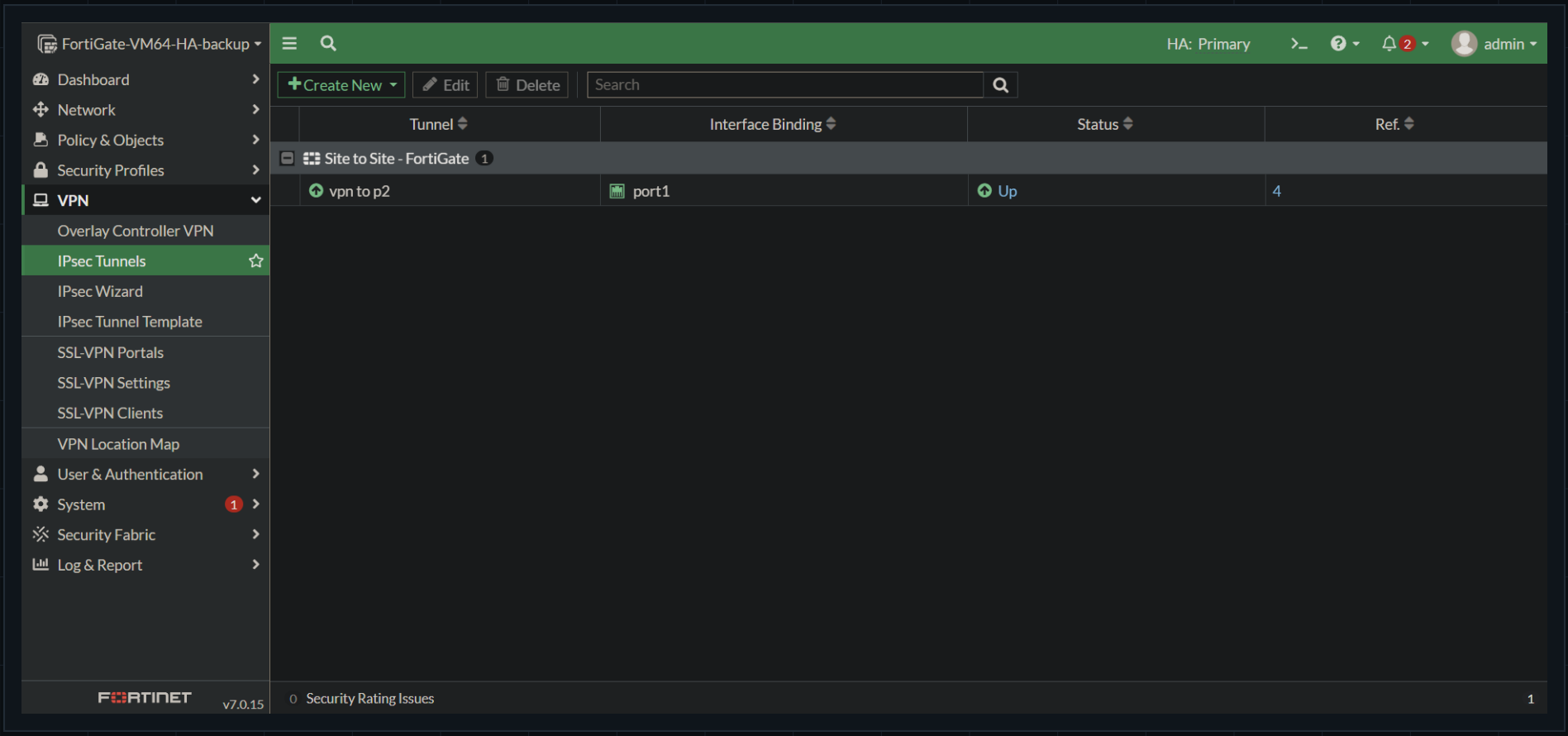


04 Firewall Policy

LAN-to-internet plus dedicated inbound/outbound rules for the VPN interface, all under UTM inspection.

SITE P1 -- HA CLUSTER

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up



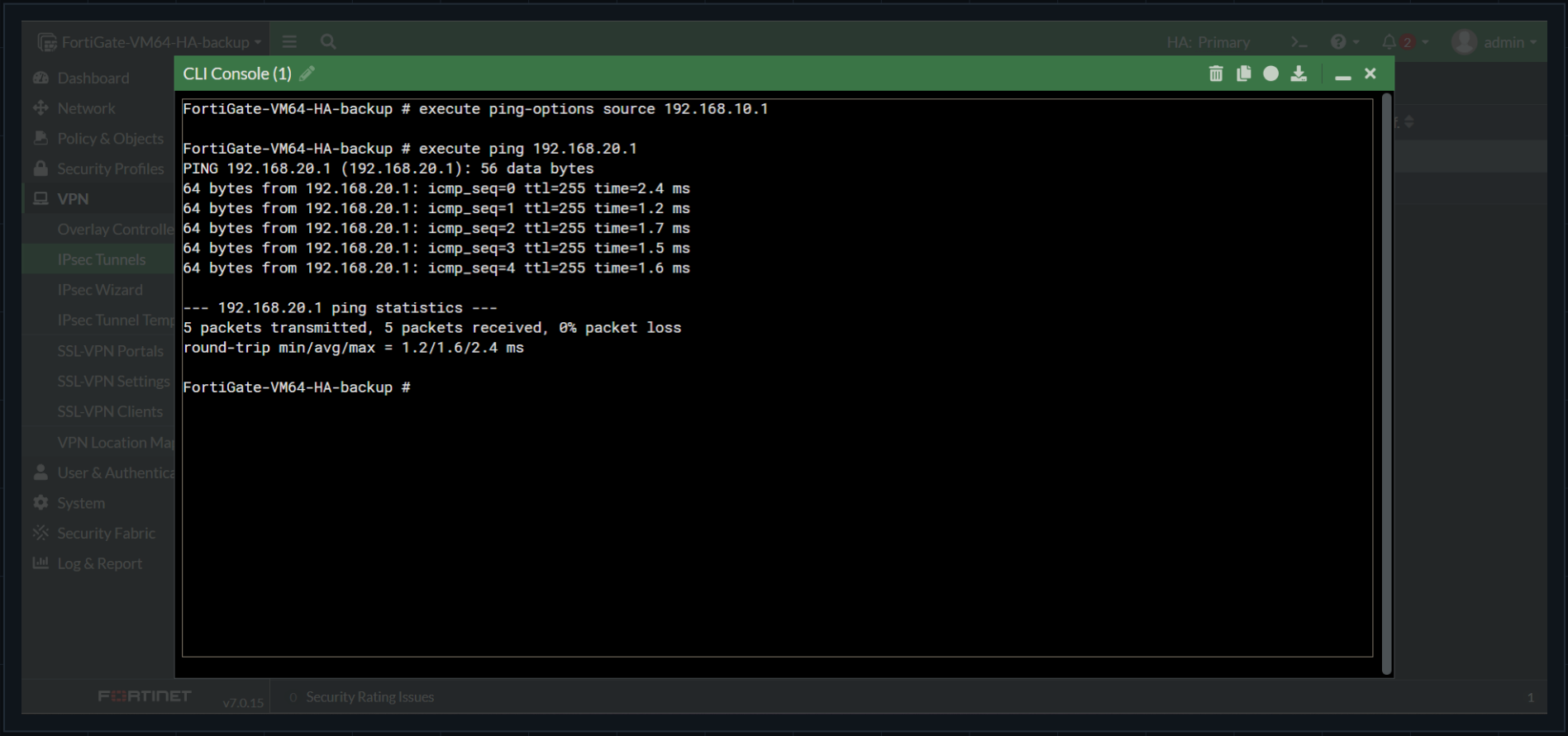
05 IPsec Tunnel

Site-to-site tunnel "vpn to p2," bound to port1, status Up.

SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up



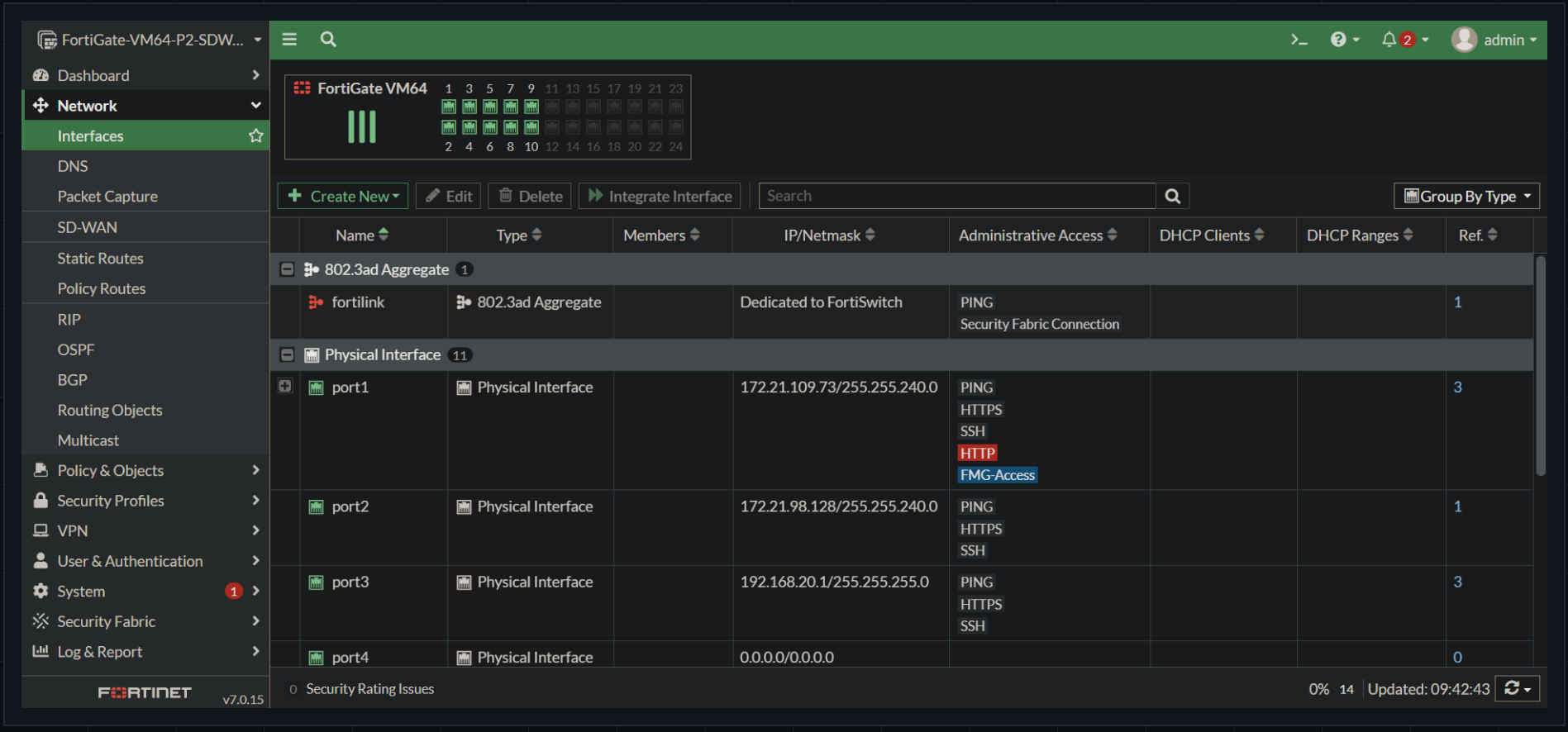
06 CLI Verification

Sourced ping from the P1 LAN to the P2 LAN (192.168.20.1) - 5/5 packets, 0% loss, 1.6 ms average.

SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up

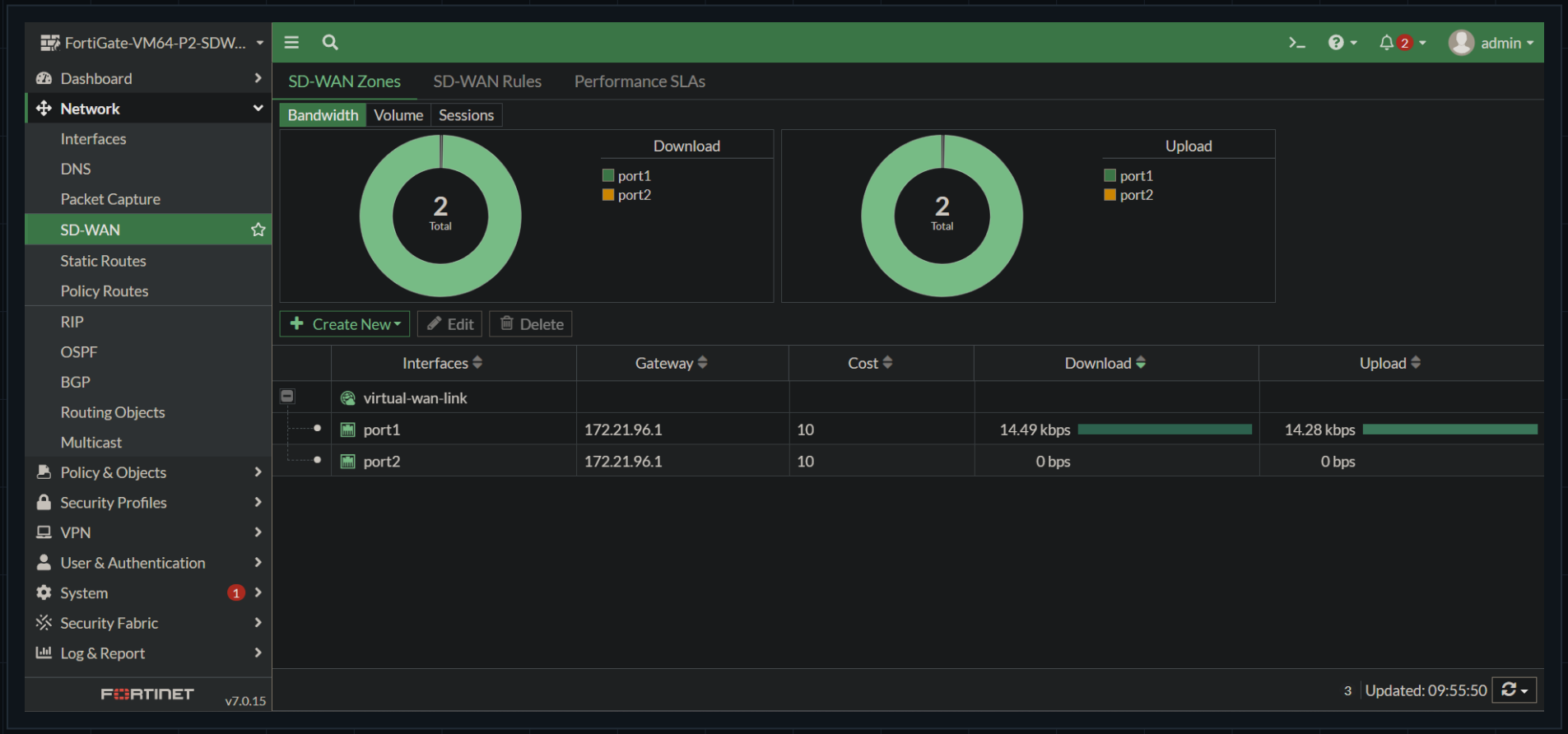


01 Interfaces

Two independent WAN links (port1, port2) plus the internal 192.168.20.0/24 LAN on port3.

SITE P2 -- SD-WAN

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up

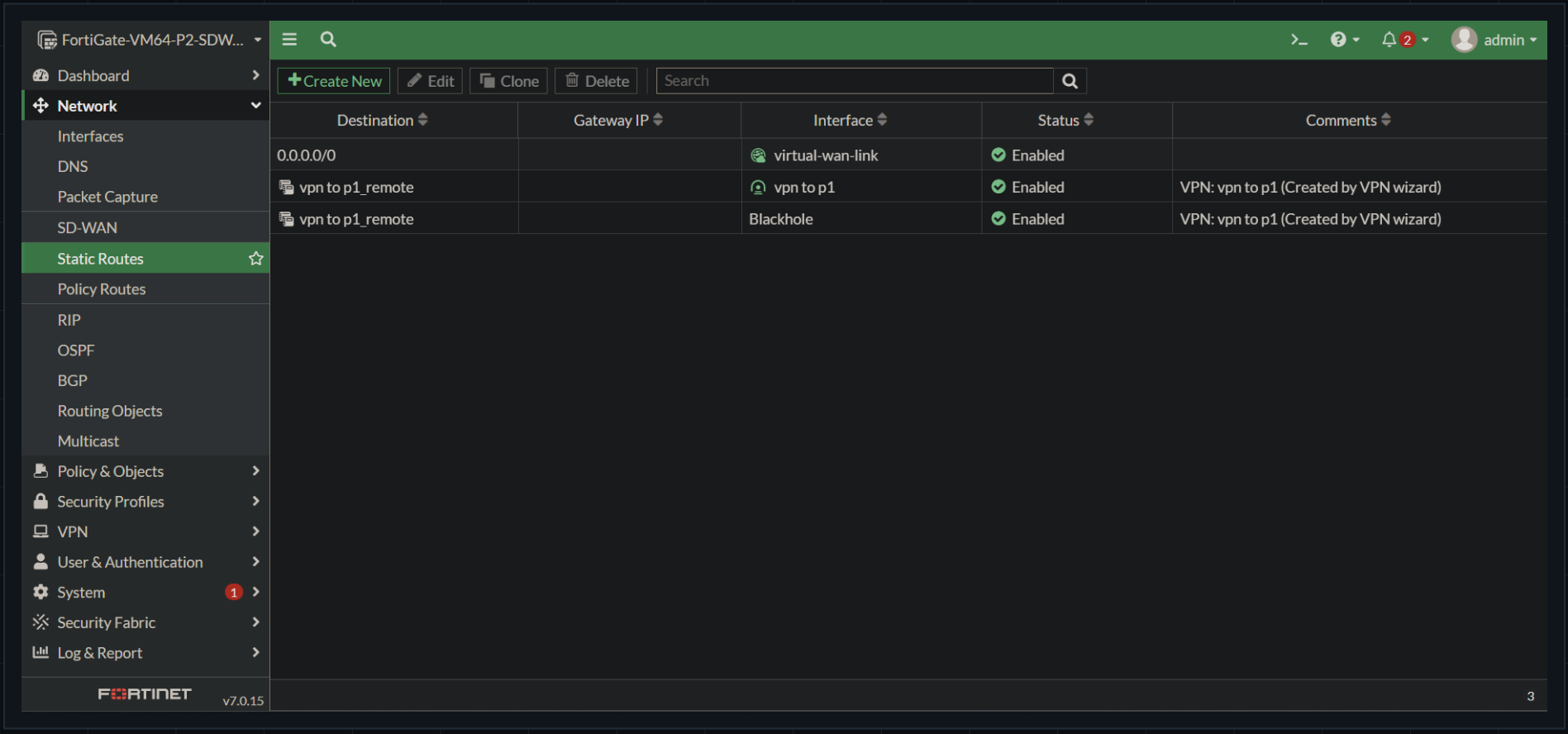


02 SD-WAN Zone

Both WAN ports grouped into the virtual-wan-link at equal cost, with live download/upload throughput per member.

SITE P2 -- SD-WAN

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up

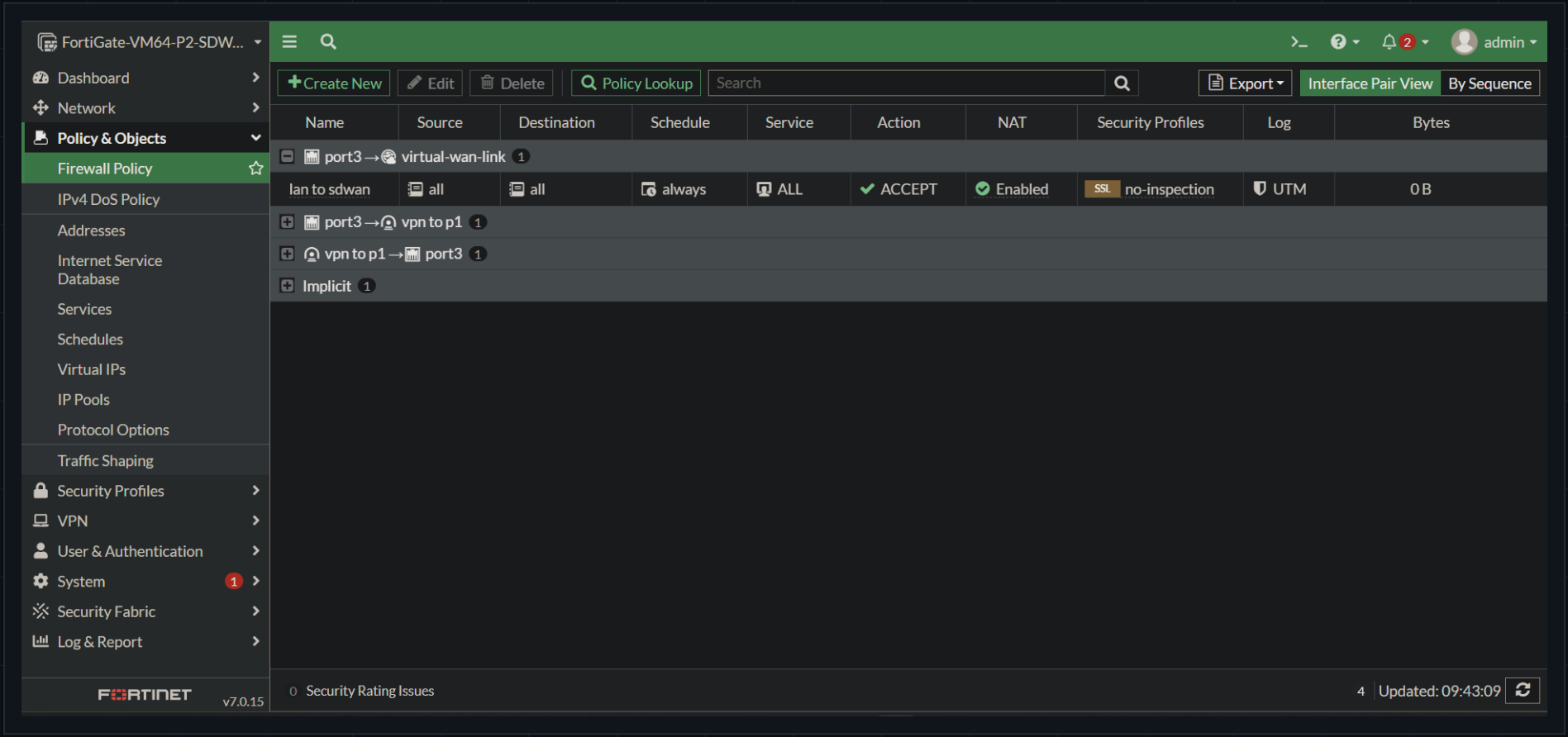


03 Static Routing

Default route rides the SD-WAN zone; the route to P1 gets the same primary-plus-blackhole failsafe pattern as P1.

SITE P2 -- SD-WAN

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up

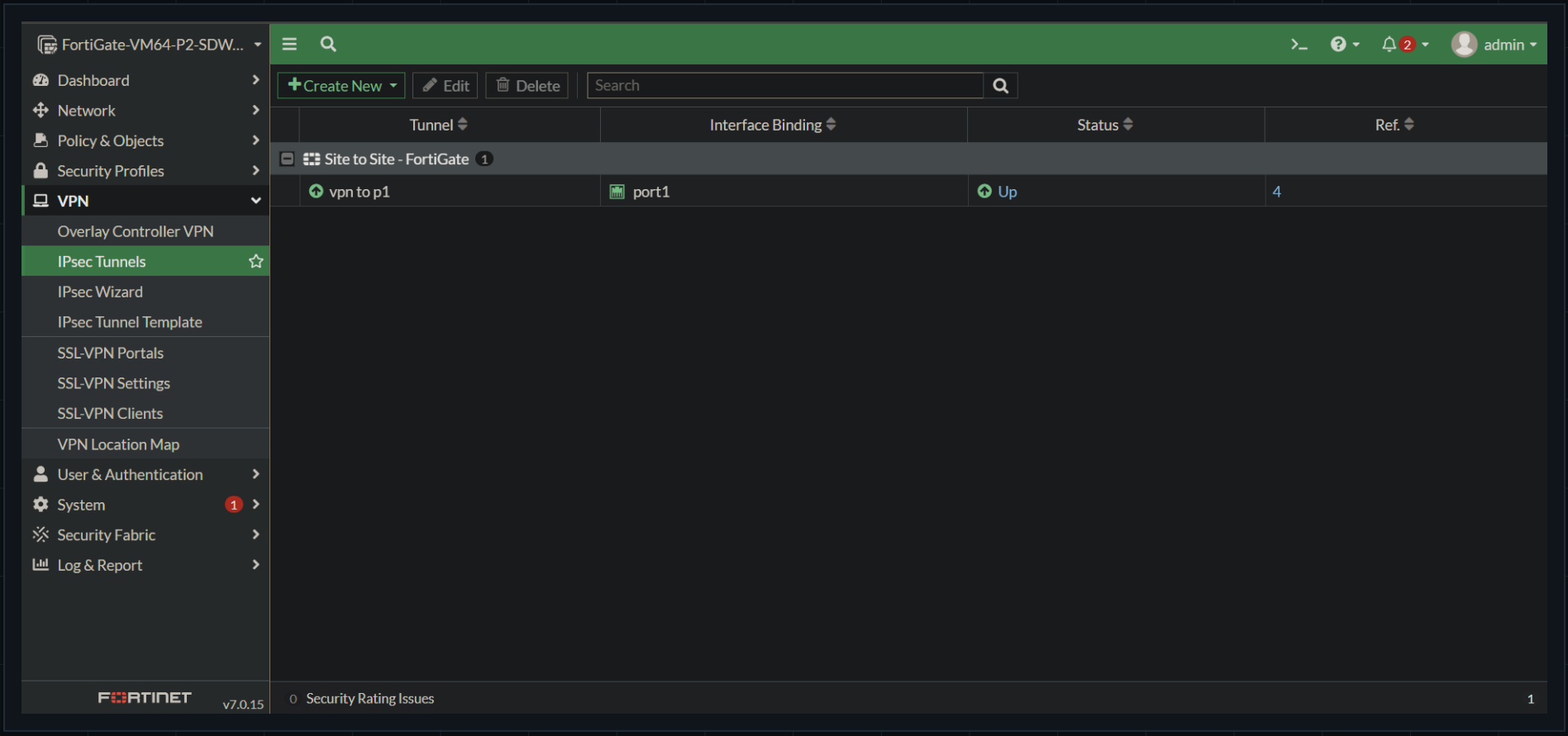


04 Firewall Policy

LAN-to-SD-WAN policy for general egress, mirrored by inbound/outbound rules for the tunnel back to P1.

SITE P2 -- SD-WAN

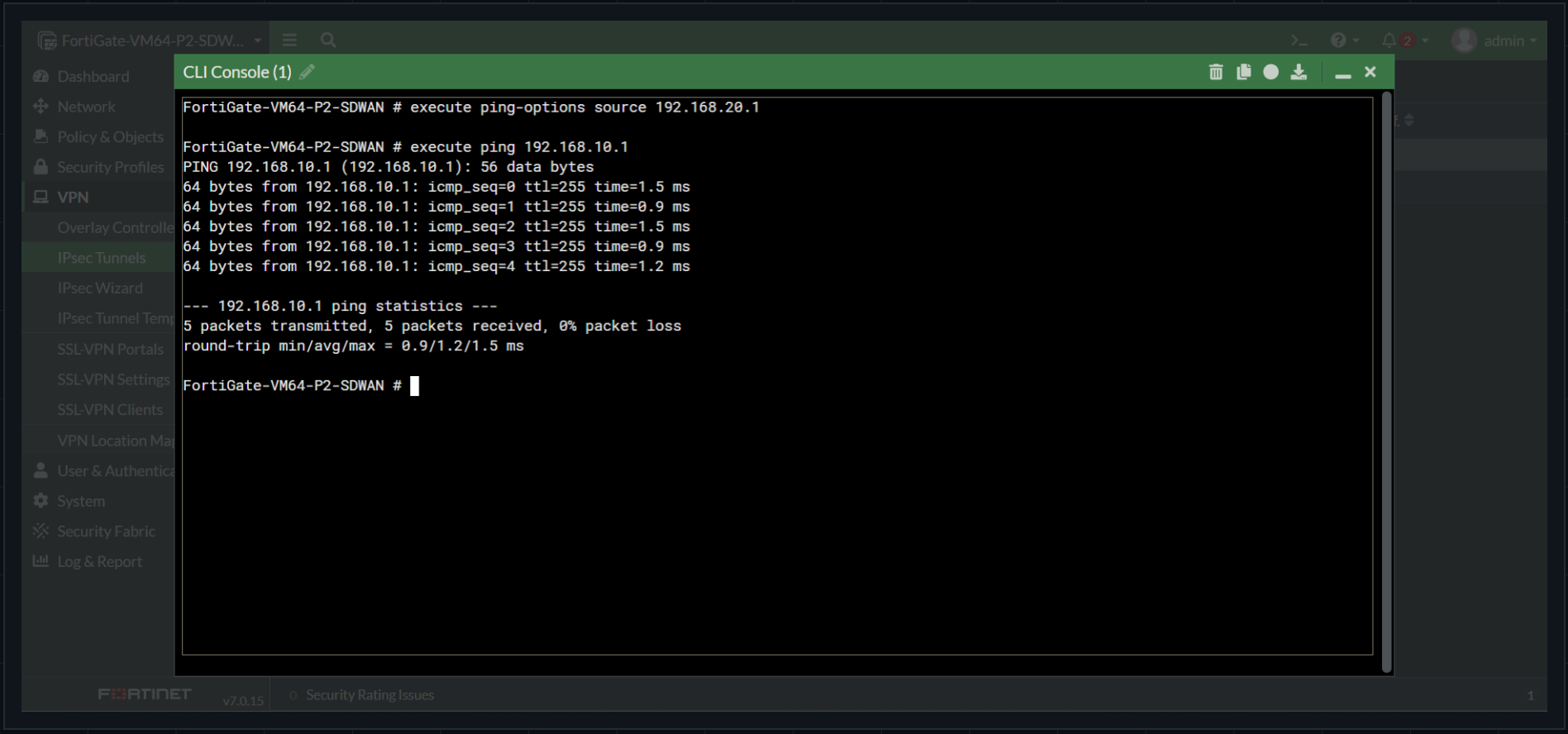
WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up



05 IPsec Tunnel

Site-to-site tunnel "vpn to p1," bound to port1, status Up.

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up



06 CLI Verification

Sourced ping from the P2 LAN to the P1 LAN (192.168.10.1) - 5/5 packets, 0% loss, 1.2 ms average, confirming bidirectional reachability.

PROJECT 02

DMZ Segmentation

A single FortiGate split into three security zones, so a compromised public-facing host in the DMZ can never pivot into the internal LAN. port1 faces the WAN, port2 hosts the internal LAN (192.168.10.0/24), and port3 is a dedicated DMZ (192.168.20.0/24) for exposed services. Policy does the isolating: DMZ-to-LAN traffic is explicitly denied, LAN-to-DMZ and LAN-to-WAN are permitted, and the implicit deny rule closes everything not named.

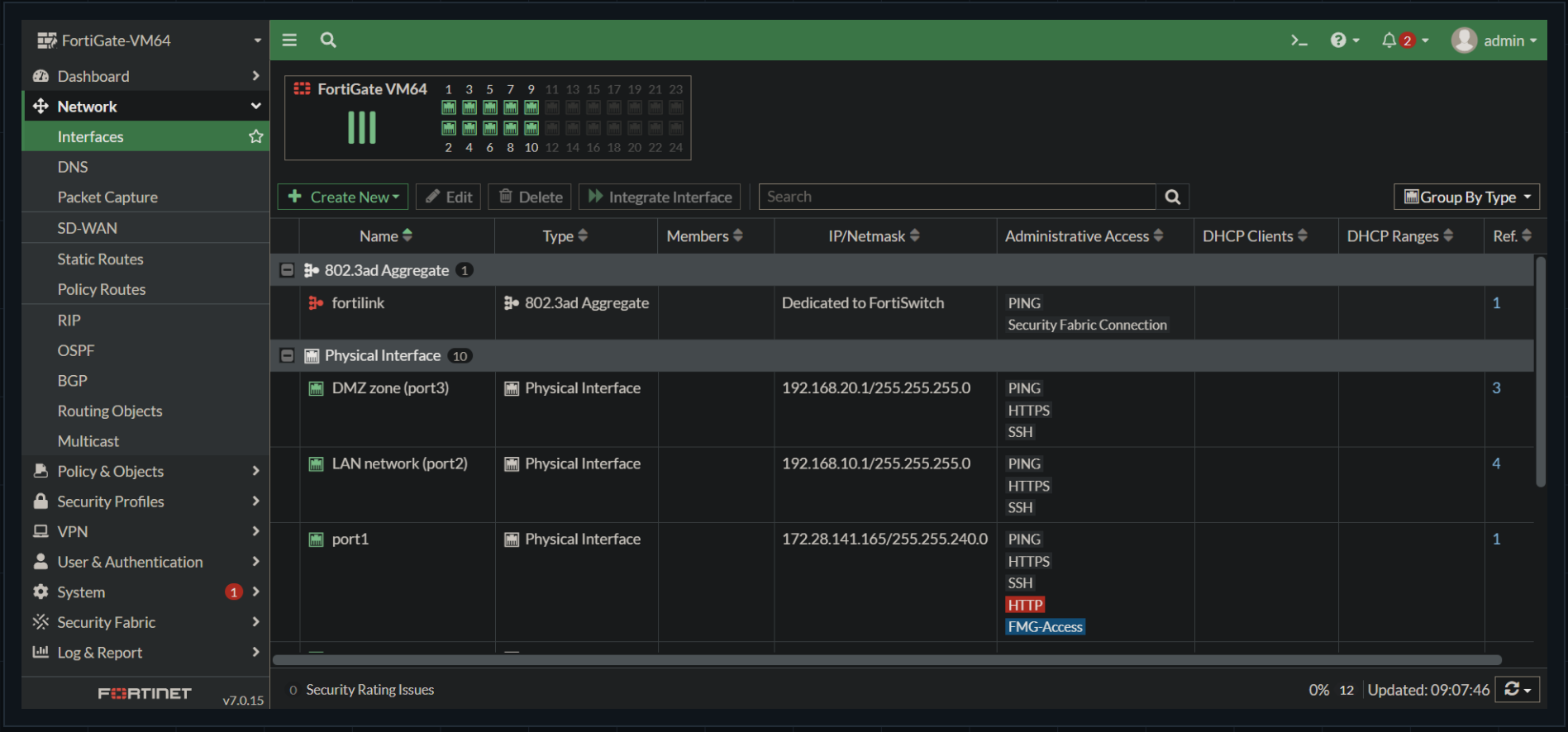
- Zones: WAN / LAN / DMZ
- DMZ -> LAN: Denied
- LAN -> DMZ: Accepted
- LAN -> WAN: Accepted

DMZ SEGMENTATION

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

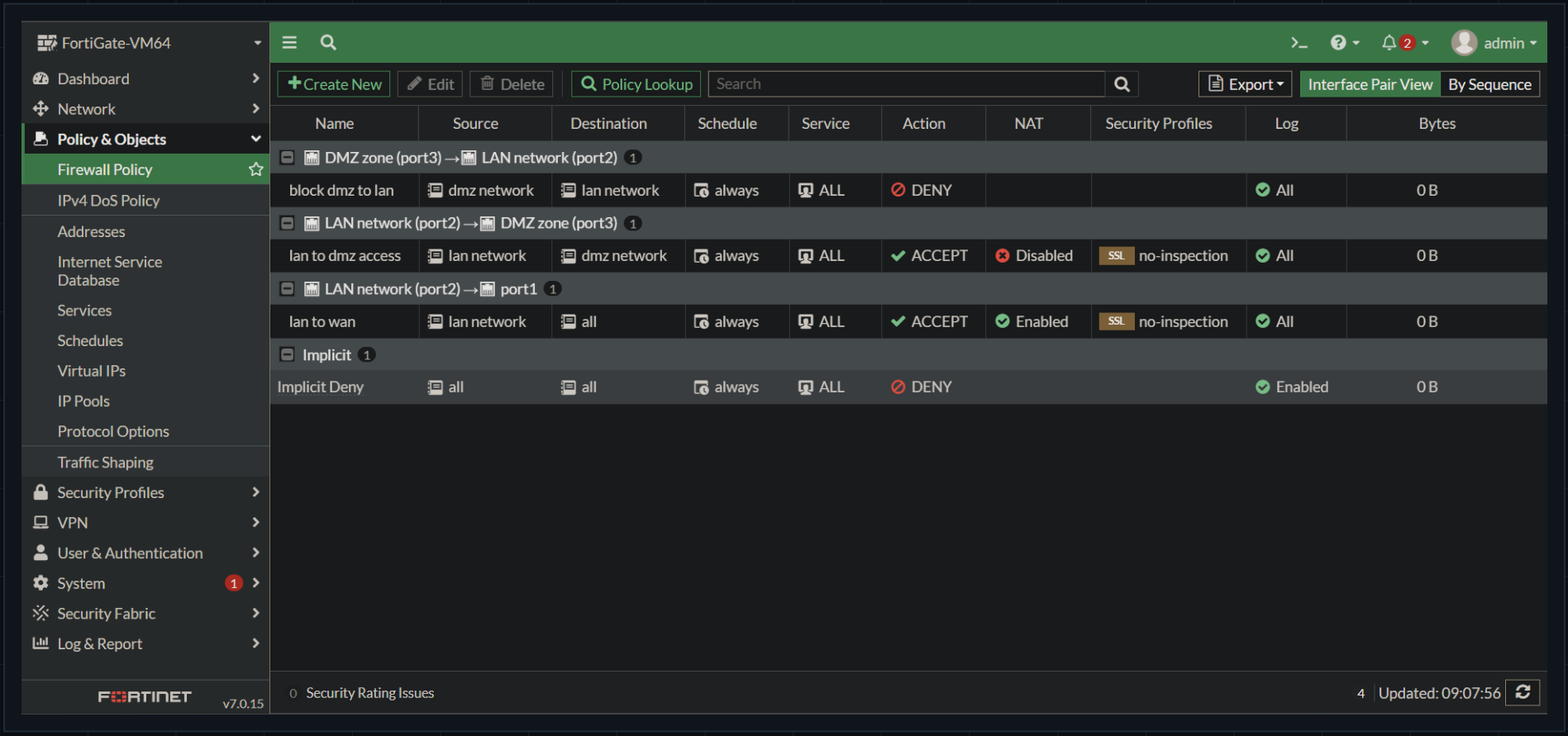


01 Interfaces

Dedicated DMZ (port3), internal LAN (port2), and WAN (port1) on a single FortiGate - each its own subnet.

DMZ SEGMENTATION

Zones: WAN / LAN / DMZ | DMZ -> LAN: Denied | LAN -> DMZ: Accepted



02 Firewall Policy

DMZ-to-LAN explicitly denied, LAN-to-DMZ and LAN-to-WAN permitted, implicit deny closes everything else.

HASSAN ISMAIL

GET IN TOUCH

Open to network & security engineering work.

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